

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

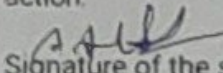
COURSE OUTCOME COVERED

C02: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

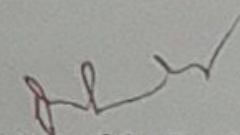
Code of Conduct:

I, A. Alhath (19252257) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%	2	1	1	2
Method Selection	20%	1	2	1	2
Solution Process	25%	2	2	2	1
Accuracy of Calculation	20%	1	1	2	2
Interpretation of Results	15%	1	1	1	1

Total Marks 29 /40


Signature of the Faculty

Step 1: Given Data

- Network Address = $192.168.20.0/24$
- Total IP addresses = 256
- Subnets:

Lab A = /26

Lab B = /27

Lab C = /28

Step 2: Calculate the Number of Hosts

Lab A (/26)

Subnet Mask: $255.255.255.192$

Host Bits = $32 - 26 = 6$

Total addresses

$$= 2^6 = 64$$

Usable Hosts

$$= 64 - 2 = 62$$

Lab B (/27)

Subnet Mask: $255.255.255.224$

Host Bits = 5

Total addresses

$$= 2^5 = 32$$

Usable Hosts

$$= 30$$

Lab C (128)

Subnet Mask: 255.255.255.240

Host Bit = 4

Total Addresses

$= 2^4 = 16$

usable Hosts

$= 14$

Step 3: Allocate the subnets

Lab A (126)

Network Address: 192.168.20.0

First Host: 192.168.20.1

Last Host: 192.168.20.62

Broadcast Address: 192.168.20.63

Lab B (127)

next available subnet starts at 192.168.20.96

• Network Address: 192.168.20.96

First Host: 192.168.20.97

Last Host: 192.168.20.110

Broadcast Address: 192.168.20.111

Lab C (128)

Final Answer:

• Lab A (126)

Network address: 192.168.20.0

Host Range: 192.168.20.1 - 192.168.20.62

Broadcast Address: 192.168.20.63 usable Hosts: 82

lab B (1/21)

Network Address: 192.168.20.0

Host Range: 192.168.20.65 - 192.168.20.94

Broadcast Address: 192.168.20.95

Usable Hosts: 30

lab C (1/22)

Network Address: 192.168.20.96

Host Range: 192.168.20.97 - 192.168.20.110

Broadcast Address: 192.168.20.111

Usable Hosts: 14

2)

A Network administrationStep 1: Given Data

- IP Address (system 1): 192.168.1.130
- IP Address (system 2): 192.168.1.190
- Subnet Mask: 255.255.255.192
- Prefix length: /2

Step 2: Calculate the subnet size.

for a /26 subnet:

- Host bits = $32 - 26 = 6$
- Total IP addresses = $2^6 = 64$
- Usable Host Addresses = $64 - 2 = 62$

Subnet increment = 64

The subnets in 192.168.1.0/24

Subnet — Network Address — Broadcast Address —

Step 3: Analyze System 1

IP Address

192.168.1.130

This address lies between 192.168.1.128 & 192.168.1.191

Therefore,

- Network Address : 192.168.1.128
- First Host : 192.168.1.129
- Last Host : 192.168.1.190
- Broadcast Address : 192.168.1.191

Result:

The IP address 192.168.1.130 is valid

Step 4: Analyze System 2

IP Address

192.168.1.190

This address also lies between 192.168.1.128 & 192.168.1.191

Therefore,

- Network Address : 192.168.1.128
- First Host : 192.168.1.129
- Last Host : 192.168.1.192
- Broadcast Address : 192.168.1.191

Result:

The IP address 192.168.1.190 is also a valid.

2.5. Usable Host Addresses.

for a /26 subnet.

• Total IP addresses = 64

• Network Address = 1

• Broadcast Address = 1

usable Host Addresses = $64 - 2 = 62$

~~Step 6: So~~

Final Answer:

• Subnet Mask: 255.255.255.192 (/26)

• Network Address: 192.168.1.128

• Valid Host Range: 192.168.1.129 - 192.168.1.190

• Broadcast Address: 192.168.1.191

• System 1 (192.168.1.130): valid host in same subnet.

• Usable Host Address: 62.

3)

Step 1: Given Data

• Network Address = 172.16.0.0 /16

• Departments:

• HR → /25

• Sale → /26

• IT → /27

• Admin → /28

Step 2: Calculate Number of Hosts.

HR (125)

- Subnet mask = 255.255.255.128
- Host Bits = 7
- Total Addresses = $2^7 = 128$
- Usable Hosts = 126

Sales (126)

- Subnet mask = 255.255.255.192
- Host Bits = 6
- Total Addresses = 64
- Usable Hosts = 62

IT (127)

- Subnet mask = 255.255.255.224
- Host Bits = 5
- Total Addresses = 32
- Usable Hosts = 30

Admin (128)

- Subnet mask = 255.255.255.240
- Host Bits = 4
- Total Addresses = 16
- Usable Hosts = 14

15: Allocate subnets

HR Department (128)

- Network Address: 172.16.0.0

- Host Range: 172.16.0.1 - 172.16.0.126

- Broadcast Address: 172.16.0.127

Sales Department (126)

172.16.0.128 available subnet.

- Network Address: 172.16.0.128

- Host Range: 172.16.0.129 - 172.16.0.190

- Broadcast Address: 172.16.0.191

17 Dept (127)

Available subnet at 172.16.0.192

- Network Address: 172.16.0.192

- Host Range: 172.16.0.193 - 172.16.0.222

- Broadcast Address: 172.16.0.223

Step 4: Remaining unused IP Address Range

after allocating E6 Admin subnet:

- Last allocated IP = 172.16.0.239

Therefore, the remaining unused addresses are:

- 172.16.0.240 - 172.16.255.255

Final Answer:

• HR (/26)

Network Address: 172.16.0.0

Host Range: 172.16.0.1 - 172.16.0.126

Broadcast Address: 172.16.0.127

Usable Hosts: 126

• sales (/26)

Network Address: 172.16.0.128

Host Range: 172.16.0.129 - 172.16.0.190

Broadcast Address: 172.16.0.191

Usable Hosts: 30

• Remaining unused IP Address Range:

172.16.0.240 - 172.16.255.255.

4)

Step 1: Given Data

• Network Address = 172.16.10.0/24

• Subnet Mask = /27

• Subnet Mask (Decimal) = 255.255.255.248

Step 2: Calculate Number of Subnets.

original prefix = /24

New prefix = /27

Borrowed Bits = $27 - 24 = 3$

Number of subnets = $2^3 = 8$

3. calculate Number

$$\text{Host Bits} = 32 - 27 = 5$$

$$\text{Total IP address per subnet} = 2^5 = 32$$

$$\text{usable Host Addressess.} = 32 - 2 = 30$$

Step 4: Determine the subnet of 172.16.10.78

The IP address 172.16.10.78 falls between.

$$172.16.10.64 \rightarrow 172.16.10.95.$$

$$\text{Subnet} : 172.16.10.64/27$$

$$\text{Network Address} : 172.16.10.64$$

$$\text{Host Range} : 172.16.10.65 - 172.16.10.94$$

$$\text{Broadcast Address} : 172.16.10.95$$

Hence 172.16.10.78 is a valid host in subnet

Step 5: Check whether belongs to same subnet.

The address 172.16.10.95 is broadcast address of subnet 172.16.10.64/27

- It belongs to same subnet

- However, it cannot be assigned to host because broadcast addresses are reserved for sending data to all devices in subnet.

Final Answer:

Network: 172.16.10.0/24

Subnet: /27

Total no. of subnet: 8

Usable Hosts per subnet: 30

Network Address: 172.16.10.64

Broadcast Address: 172.16.10.95

5)

A network uses the IPv6 address:

Step 1: Given Data

IPv6 Address:

2001:0db8:0000:0000:0000:8f00:00
42:8329

prefix length: /64

Step 2: Compress the IPv6 Address

IPv6 Compression Rules.

- i) Remove leading zeros from each block.
- ii) Replace one continuous sequence of all-zero blocks with '::' (only once)

Step 3:

Applying the Rules

Original Address:

2001:0db8:0000:0000:0000:8f00:0042:8329

compress the consecutive zero blocks:
compressed IPv6 Address

2001:db8:aff00:42:8329

Step 4: Determine the network prefix (/64)

Therefore

Network Prefix

2001:db8:0000:0000::/64

or in compressed form:

2001:db8::/64

Step 5: Calculate the Number of Hosts.

IPv6 Address Length = 128 bits.

Network prefix = 64 bits

Host Bits = $128 - 64 = 64$

Total = 2^{64}

= 18,446,744,073,709,551,616.